01 — Executive Summary

Blood Bank is a mobile application built for Libya that connects blood donors directly with hospitals and blood banks during emergencies. It is the **first platform of its kind in Libya**, addressing a critical gap in emergency healthcare coordination. The app is fully functional, tested, and ready for real-world deployment pending partnership with a licensed medical institution.

2	8	3	0
User Types	**Core Screens**	**Urgency Levels**	**Patient Data**
Donor & Medical Clinic	Fully designed & working	Critical / Urgent / Moderate	No medical records stored

Why This Matters

Libya currently has no mobile platform that allows blood banks or hospitals to broadcast emergency blood requests to pre-registered donors. When a patient needs O- blood urgently, staff must call family members, post on Facebook, or rely entirely on whoever happens to be nearby. Blood Bank replaces this broken process with an instant, location-aware notification system that reaches the right donors in seconds.

Critically, the app stores **no patient data**. Hospitals use it only to broadcast a request (blood type, urgency, location, phone number). Donors respond voluntarily. This design removes data privacy concerns entirely and makes institutional adoption straightforward.

02 — App Overview & Core Purpose

What the App Does

Blood Bank is a React Native (Expo) mobile application with two distinct user sides: one for blood donors and one for medical clinics and blood banks. The two sides work together to create a real-time emergency response network.

Medical Clinic Posts emergency request (blood type, urgency, units needed) → **App Server** Matches by blood type and city instantly → **Donors Notified** See request, call clinic or tap 'I Can Donate'

Core flow: clinic posts request → server matches → donors receive alert

Design Principles

• No patient data stored — zero privacy risk • City-based matching — notifications reach the right donors • Urgency levels — Critical, Urgent, Moderate • Health screening at signup — only eligible donors registered

• Libyan cities pre-loaded — Tripoli, Benghazi, Misrata, Zawia + more • License number required for medical clinics — built-in trust signal • Dark, accessible UI — tested on Android • Donation history tracked per donor

03 — User Types & Flows

User Type A: Blood Donor

Step 1	**Role Selection** Opens app → taps 'Blood Donor'
Step 2	**Basic Info** Name, email, phone, blood type, city, password
Step 3	**Health Screening** Age verification (18-65), medical condition checklist (HIV, Hepatitis B/C, Cancer, TB, Malaria, Heart Disease)
Step 4	**Eligibility Result** Green screen if eligible, amber warning if not — with compassionate messaging
Step 5	**Home Dashboard** Sees their blood type, city, eligibility status, and live emergency requests
Step 6	**Emergency Response** Taps 'I Can Donate' or calls the clinic directly from the alert card
Step 7	**Profile** Views donation history, personal info, eligibility status

User Type B: Medical Clinic / Blood Bank

Step 1	**Role Selection** Opens app → taps 'Medical Clinic'
Step 2	**Registration** Clinic name, email, phone, address, city, license number (MOH format), password
Step 3	**Clinic Portal** Dashboard showing Active / Fulfilled / Total requests at a glance
Step 4	**Broadcast Request** Selects blood type, urgency level, units needed, adds notes — taps Broadcast
Step 5	**Confirmation** Instant confirmation that alert was sent to matching donors in that city
Step 6	**Clinic Profile** Shows license badge, contact info, request stats

04 — Feature Inventory

Donor Side Features

Feature	Status	Notes
Role selection screen	✓ Done	Clean onboarding entry point
2-step donor registration	✓ Done	Basic info + health screening in one flow
Blood type selector (all 8 types)	✓ Done	A+, A-, B+, B-, AB+, AB-, O+, O-
Libyan city selector	✓ Done	Tripoli, Benghazi, Misrata, Zawia + more
Health condition screening	✓ Done	HIV, Hepatitis B/C, Cancer, TB, Malaria, Heart Disease
Eligibility result screen	✓ Done	Eligible (green) and Not Eligible (amber) states
Donor home dashboard	✓ Done	Blood type, city, eligibility, emergency feed
Emergency alert feed	✓ Done	Real-time list with blood type badges and urgency tags
Blood type filter tabs	✓ Done	Filter alerts by blood type on Alerts screen
'I Can Donate' action	✓ Done	One-tap response to an emergency request
Direct call button on alerts	✓ Done	Phone number of clinic shown on each card
Donation history	✓ Done	Shows past donations with date, clinic, and units
Donor profile screen	✓ Done	Name, email, phone, city, blood type, eligibility status
Arabic language support	■ Planned	RTL layout for wider Libyan adoption
WhatsApp share button on alerts	■ Planned	Share emergency to contacts
Push notification when offline	→ Next	Alert even when app is closed

Medical Clinic Side Features

Feature	Status	Notes
Clinic registration with license number	✓ Done	MOH license field for trust verification
Clinic portal dashboard	✓ Done	Active / Fulfilled / Total request counters
Emergency broadcast form	✓ Done	Blood type, urgency, units, additional notes
Urgency level system	✓ Done	Critical (immediate), Urgent (24h), Moderate (48-72h)
Broadcast confirmation dialog	✓ Done	Confirms city + blood type that was alerted
Clinic profile screen	✓ Done	Shows 'Licensed' badge, contact info, stats
Request management (Requests tab)	✓ Done	View all sent requests
Fulfilled request tracking	■ Planned	Mark a request as fulfilled when blood is received
Donor response visibility	■ Planned	See how many donors responded to each alert
Clinic analytics dashboard	■ Planned	Response rates, peak times, blood type demand

05 — Technical Architecture

Technology Stack

Layer	Technology	Purpose
Frontend	React Native (Expo)	Cross-platform mobile app (iOS + Android)
Language	JavaScript / TypeScript	App logic and UI components
Navigation	Expo Router / React Navigation	Screen navigation and deep linking
Notifications	Expo Notifications	Push alerts to donor devices
Backend	To be confirmed	API for auth, data, and broadcast logic
Database	To be confirmed	Donor profiles, clinic accounts, request logs
Auth	Email + Password	Separate auth flows per user type

Data Privacy Architecture

No patient data is stored or transmitted. The app only stores: donor name, contact info, blood type, city, and self-reported health eligibility screening results. Medical clinics store: clinic name, address, license number, contact info. Emergency requests contain only: blood type needed, urgency level, units needed, and optional text notes. No patient names, diagnoses, or medical records are ever collected, stored, or shared.

Notification Flow

When a clinic broadcasts an emergency request, the system identifies all registered donors in the same city with the matching blood type who are currently marked as eligible. A push notification is sent to each matched donor's device. The donor sees the clinic name, location, urgency level, and a direct call button — they do not see any patient information. The donor can tap 'I Can Donate' to signal intent or call the clinic directly.

06 — MVP Scope vs Future Roadmap

What the MVP Delivers (Now)

The current version of Blood Bank is a complete, functional MVP. Everything listed below is built, tested on device, and ready for real users:

• Full donor registration with health screening • Full clinic registration with license number • Emergency broadcast system • Real-time alert feed for donors • Blood type filtering on alerts • Urgency level classification (3 tiers) • Direct call-to-clinic from alert cards

• Donation history tracking • Donor eligibility status tracking • Clinic portal with request statistics • Broadcast confirmation flow • Donor and clinic profile screens • City-based donor matching • Ineligibility screen with compassionate UX

Phase 2 — Partnership Launch

• Arabic (RTL) language support • WhatsApp share button on alerts • Push notifications when app is closed • Donor response confirmation to clinics

• Fulfilled request marking • Clinic analytics dashboard • Google Play Store public release • Verified clinic badge system

Phase 3 — Scale & Growth

• Multi-city expansion beyond Tripoli • Regular donation scheduling / reminders • Donor leaderboard and recognition

• Blood type compatibility matching • API for hospitals to integrate directly • iOS App Store release

07 — Partnership Model

How We Work with Blood Banks

Blood Bank is not a competitor to blood banks or hospitals — it is a tool that makes their work faster and more effective. The partnership model is simple:

1	One trusted blood bank partners with us first as the anchor institution in Tripoli.
2	The blood bank's staff register through the Medical Clinic portal using their official license number.
3	They begin using the Broadcast Emergency Request feature for real blood needs.
4	Donors in Tripoli register organically via word of mouth, social media, and the blood bank's own channels.
5	As the network grows, other hospitals and clinics in Tripoli can join and broadcast their own requests.
6	We expand city by city — Benghazi, Misrata, Zawia — as we gain local partners in each city.

What We Ask From a Partner Blood Bank

- Official clinic name and license number
- Contact phone number for donors to call
- Physical address for navigation
- One staff member to manage app broadcasts
- Willingness to share the app with patients and community
- Feedback on real usage to help us improve
- No cost — the app is free for partner institutions

This is a free tool. There is no cost for blood banks or hospitals to use Blood Bank. We are building this because Libya needs it. Our only ask is your participation and your trust.

08 — How to Get Involved

Try the App Right Now

Blood Bank is available for immediate testing on any Android device via Expo Go. No app store submission required — you can see the full app working in under 2 minutes:

Install	Download 'Expo Go' from the Google Play Store on any Android phone
Scan	Scan the QR code provided by our team — the app opens instantly
Explore	Try registering as a donor, or log in as a clinic and broadcast a test request
Ask	We are available to walk through any screen or answer questions in person or over a call

Next Steps

Short term	Medium term	Long term
• Connect with first blood bank partner in Tripoli • Gather real user feedback • Deploy push notifications	• Arabic UI support • Google Play public release • Expand to Benghazi	• Multi-city network • Hospital API integration • iOS release

Blood Bank was designed and built in Libya, for Libya. Every feature reflects a real need in our healthcare system. We look forward to working with institutions that share our commitment to saving lives.